

Mapping the potential public and private returns to Enhanced Rock Weathering in Sub-Saharan Africa

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Abstract

Enhanced Rock Weathering (ERW) – spreading finely ground silicate rock such as basalt on cropland – sits at an unusual intersection of two value streams: it durably removes atmospheric CO₂ (a public good, monetizable through voluntary carbon markets, VCM) and it neutralizes soil acidity much like agricultural lime, raising yields (a private return to the farmer). We develop a spatially explicit, decomposed *ex ante* profitability model for 44 Sub-Saharan African (SSA) countries and 23 crops that separates the per-hectare gross margin of basalt-ERW into an agronomic (private) term and a carbon (public) term, each net of the full delivered cost. We map three decision-relevant envelopes: land where the private return alone justifies investment, land where the public return alone justifies it, and their intersection. Crediting only durably exported carbon – deducting the alkalinity consumed neutralizing soil acidity, which re-releases its CO₂ just as lime does – we evaluate the public (carbon) and private (yield) cases together in the equilibrium steady state. Under a spatially targeted lime-equivalent

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dose the private envelope (4.3 Mha) exceeds the durable public envelope (2.0 Mha), and the two coincide on 1.3 Mha; but the *combined* envelope is far larger (7.6 Mha), because on roughly 2.7 Mha neither stream alone covers the cost yet together they do. This is the central economic result: voluntary carbon finance is not a bonus on an already-profitable practice but the margin that makes an otherwise-unprofitable agromonic investment viable. Properly accounting for acidity neutralization cuts credited removal by roughly 70% at first application and shows that durable removal is largely a steady-state phenomenon. It also implies that the lime-requirement dose is CDR-minimal by construction, so the durable carbon case must be diagnosed at doses above the agronomic optimum: on a uniform 20 t ha⁻¹ dose the public envelope is 4.0 Mha carrying 20.4 Mt of durable removal, against 2.0 Mha and 5.6 Mt at the targeted dose. The carbon case is bound by the realized CO₂ removed per tonne of rock (area-weighted median marginal abatement cost \$276/t CO₂ on that uniform dose, \$330 on the targeted dose), the largest and least-resolved lever. We map a robust core-targeting geography and the field-data priorities that would most improve it.

Keywords: enhanced rock weathering; carbon dioxide removal; voluntary carbon markets; soil acidity; agricultural liming; Sub-Saharan Africa; additionality; *ex ante* analysis.

JEL: Q12, Q15, Q54, Q56, O13.

1 Introduction

Enhanced rock weathering (ERW) has attracted considerable attention as a carbon dioxide removal (CDR) pathway that promises climate mitigation while potentially delivering material benefits to farmers (Beerling et al., 2020, 2025b; Suhrhoff et al., 2026). The mechanism is well understood. Silicate rock – most commonly basalt – is quarried, ground to a fine dust, and spread on agricultural fields. There, rainwater and the CO₂ respired by roots and soil microbes form carbonic acid, which dissolves the silicate minerals; this reaction consumes CO₂

and converts it into dissolved bicarbonate (HCO_3^-) while releasing base cations such as Ca^{2+} and Mg^{2+} . The bicarbonate is carried by drainage water through rivers to the ocean, where the captured carbon is stored on geological timescales (in excess of 10,000 years) (Beerling et al., 2020; Renforth and Henderson, 2017). The same reaction that sequesters carbon also neutralizes soil acidity: the released cations raise soil pH and base saturation, relieving the aluminum and manganese toxicity and nutrient-availability constraints that depress yields on acid soils, so that ERW can raise crop yields much as agricultural lime does (Beerling et al., 2024; Skov et al., 2024; Haque et al., 2025).

While the mechanisms are well established, how ERW would play out in practice – particularly in data-limited geographies – is far less substantiated. In particular, the economics that underlie return-on-investment calculations have not been evaluated in a spatially explicit, granular way, which makes it difficult to know how and where to invest. This gap matters because the public investment case, tied to carbon removal, may be spatially distinct from the private case – the locations where the value of yield gains exceeds the cost of applying ERW. Where these two return surfaces diverge, alternative funding strategies – voluntary carbon market (VCM) financing, private input-market development, or some blend of the two – face very different spatial opportunity sets. The absence of explicit, comparable estimates is thus a barrier to investment and targeting. One concrete version of the targeting problem is the question a carbon-market investor would ask: where can payments for carbon sequestration also deliver benefits to smallholder farmers?

In this paper we take a spatially explicit, *ex ante* modeling approach to characterize the benefit streams that ERW would generate across different geographical contexts in Sub-Saharan Africa (SSA), an understudied region with large potential payoffs to the practice. Our approach is grounded in the observation that the economics of ERW are fundamentally governed by spatial conditioners: CDR potential depends on soil chemistry and rainfall; the yield dividend from acidity neutralization varies with crop, soil type, and baseline acidity; and the delivered cost of feedstock to the farm gate is built up from quarry proximity,

grinding energy, and transport distance. Bringing these spatially varying factors into a single accounting framework, we evaluate how the distributions of public and private returns compare with one another and with the site-specific delivered cost of ERW.

Our work draws on and connects two largely separate literatures, reviewed in Section 2: a fast-growing body that substantiates the geochemistry and carbon-removal potential of ERW but rarely examines its on-farm economics, and a spatially explicit agronomic and economic literature on the returns to soil-acidity remediation through liming. The few ERW yield trials in Sub-Saharan Africa document agronomic responses — including multi-season absolute yield gains in Ghanaian maize (Oppong Danso et al., 2025; Akortey et al., 2026) and a smallholder maize trial in Kenya (Haque et al., 2025) — but stop short of spatially explicit economics, while the liming literature (Silva et al., 2025; Hijbeek et al., 2021) supplies the integrated, spatially explicit framework we adapt.

Our value-added is to take that liming framework and reconfigure it for ERW, accounting jointly for the agronomic yield benefit and the carbon-removal benefit, net of the spatially varying delivered cost of feedstock, while operationalizing the mechanistic CDR understandings from the geochemical literature at high spatial resolution. To our knowledge this is the first analysis to map the private and public benefit streams of ERW side by side and net of location-specific delivery costs. Doing so lets us delineate the opportunity space for carbon-market financing, which is especially attractive where it also delivers smallholder co-benefits, and ask where the private return alone is large enough that a market for ERW could be built on farmers’ own willingness to pay.

We address six questions, the first three scoping the financing opportunity and the last three turning the estimates into decisions and a model-improvement agenda.

1. Where is there a case for a *private market* for ERW, regardless of public returns — that is, where do yield-based returns alone exceed the delivered cost, so that farmers or input suppliers could fund deployment without carbon finance?
2. Where is there a case for *voluntary-carbon-market (VCM) investment*, regardless of

private returns – where does the value of durable carbon removal alone exceed the delivered cost?

3. Where do the two *coincide*, strengthening the proposition for a carbon buyer – where a purchased tonne of removal also delivers a smallholder yield co-benefit?

4. How robust are these answers to parameter uncertainty (delivered cost, carbon price, yield response, CDR rate, and the carbon-accounting choice)?

5. Given that uncertainty, which areas should early *pilots* target?

6. Which assumptions and processes – and in which areas – would targeted *field data* most improve, to drive an iterative modeling loop?

A point on framing: the private return and the carbon-market *co-benefit* are the same quantity – the economic value of ERW-induced yield gains – seen from two vantage points, the smallholder who captures it directly and the carbon buyer who values it as bundled impact. Questions 1–3 map onto the private, public, and intersection return surfaces we estimate.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature in more detail. Section 3 sets out our conceptual framework and modeling approach. Section 4 presents our core results and the sensitivity of our conclusions to parameter uncertainty. Section 5 discusses how the results may inform investment strategies, pilot-program targeting, and further research, and Section 6 offers concluding remarks.

2 Background and related literature

ERW as a joint product. Enhanced rock weathering is distinctive among carbon-removal options in producing two benefits from a single action. When crushed silicate rock, typically basalt, dissolves in soil, the reaction consumes carbonic acid and converts CO₂ into bicarbonate while releasing base cations such as Ca²⁺ and Mg²⁺ (Beerling et al., 2020; Kantola

et al., 2017). The bicarbonate, once it leaches to rivers and the ocean, stores carbon on geological timescales; the same cations raise soil pH and relieve the aluminum and manganese toxicity that depress yields on acid soils, so the agronomic effect closely parallels agricultural lime (Swoboda et al., 2022; van Straaten, 2002). The contrast with lime is instructive, since lime’s alkalinity carries carbon that is re-released when it neutralizes acidity, making liming carbon-neutral at best, whereas the silicate’s alkalinity is generated by consuming atmospheric CO₂. Field trials increasingly document the yield side of this joint product, in temperate systems (Skov et al., 2024; Beerling et al., 2024) and, in Sub-Saharan Africa, in Ghanaian and Kenyan maize amended with fine-grained silicate rock flour (Oppong Danso et al., 2025; Akortey et al., 2026; Haque et al., 2025). It is this coupling of a public good and a private good in one practice that motivates analyzing the two returns together.

CDR potential and cost. A large literature estimates the carbon-removal potential and cost of ERW. Early techno-economic work derived global potentials and \$/t CO₂ cost curves and identified grinding energy and feedstock chemistry as the dominant cost and efficiency levers (Strefler et al., 2018), and continental assessments place croplands at the center of feasible deployment (Beerling et al., 2020, 2025a). Subsequent modeling has sharpened how climate governs weathering rates and hence the spatial distribution of potential (Baek et al., 2023; Deng et al., 2023). A parallel and still unsettled strand concerns how much of the gross weathering signal is durable: bicarbonate can re-precipitate as pedogenic carbonate or fail to export, so realized removal can fall well below stoichiometric ceilings (Renforth, 2019; Kanzaki et al., 2023), and recent reviews stress the resulting quantification and permanence uncertainties and the monitoring needed to resolve them (Schiedung et al., 2026; Levy et al., 2024; Beerling et al., 2025b; Suhrhoff et al., 2026). This body of work is rich on the geochemistry but typically frames ERW as a carbon supply curve, abstracting from the farm economy in which deployment must actually occur.

Table 1: Basalt application rates in comparable ERW assessments.

Study	Scope	Application rate
Baek et al. (2023)	Global croplands (model)	10 t/ha/yr
Tu et al. (2026)	Global adoption/equity (model)	10 t/ha
Beerling et al. (2020)	Global croplands (model)	40 t/ha/yr
Beerling et al. (2025a)	US croplands (model)	40 t/ha/yr
Kroeger et al. (2026)	Life-cycle / techno-economic	50 t/ha
Skov et al. (2024)	Temperate field trial (oat)	18.9 t/ha
Oppong Danso et al. (2025)	Ghana field trial (maize)	10, 50 t/ha
Akortey et al. (2026)	Ghana field trial (maize)	50, 100 t/ha
Jordan et al. (2026)	India on-farm (rice)	25–50 t/ha
This study	SSA, spatial (private + public)	targeted (lime req.) + uniform 10/20/50 t/ha

Application rates in comparable assessments. Spatially explicit assessments of ERW on croplands have almost universally applied a *uniform* basalt rate, and those rates cluster in a narrow band (Table 1). Global and regional models use 10 to 50 t/ha, often annually: 10 t/ha/yr in Baek et al. (2023) and Tu et al. (2026), 40 t/ha/yr in Beerling et al. (2020) and Beerling et al. (2025a), and 50 t/ha in the life-cycle and techno-economic assessment of Kroeger et al. (2026). Field trials test a wider range, from ~ 19 t/ha in a temperate trial (Skov et al., 2024) and 10 to 50 t/ha in West African experiments (Oppong Danso et al., 2025) to 50 to 100 t/ha (Akortey et al., 2026), and on-farm smallholder deployments in India apply 25 to 50 t/ha with documented yield and income gains (Jordan et al., 2026); reviews note experimental rates as high as several hundred t/ha (Schiedung et al., 2026), and a systematic compilation across 98 experimental studies reports a field-trial median of 20 t/ha (interquartile range roughly 6–100) (Suhrhoff et al., 2026). Our uniform scenarios (10/20/50 t/ha) therefore span the modeling mainstream, with our maximum at its upper edge and well below extreme experimental doses. Our targeted allocation departs from this literature: instead of a fixed uniform rate, it sizes the dose to each soil’s lime requirement, which is precisely what couples the carbon and agronomic accounts and motivates the net-export correction of Section 3.4.

Soil acidity and the economics of liming in SSA. Because ERW is, agronomically, “liming with a different rock,” the soil-acidity-remediation literature provides our closest methodological template. Acid soils are widespread across SSA and impose large, quantifiable yield penalties; recent continental work estimates that acidity depresses yields on 32.7 Mha, about 23% of cropland, a roughly US\$6 billion burden, a substantial share of which could be profitably alleviated by liming on the order of 6 to 9 Mha (Silva et al., 2025). That assessment builds on spatially explicit lime-requirement models (Aramburu Merlos et al., 2023) and connects to farm-level analyses of the multi-year economics and greenhouse-gas trade-offs of liming, for example in western Kenya (Hijbeek et al., 2021). We adapt this integrated, spatially explicit framework, reconfiguring its costs and crop-response functions for basalt and adding the carbon-removal benefit and the delivered-cost geography that ERW introduces.

Adoption and smallholder finance. Whether any of this potential is realized depends on who can finance it. The defining feature of the ERW investment, as with liming, is a large upfront outlay (on the order of \$600/ha for basalt) against benefits that accrue over several seasons. For credit-constrained smallholders with short decision horizons, such front-loaded investments are precisely those that the technology-adoption literature identifies as systematically under-adopted absent finance, insurance, or subsidy (Jack, 2013). This makes the locus of returns a first-order question: where farmers themselves capture enough value to pay, deployment can proceed through input markets or cost-sharing arrangements; where they do not, it requires external finance. Estimating the private return is therefore not only of academic interest but directly informs which financing instrument is feasible where.

Voluntary carbon markets and additionality. The most discussed external financing channel for ERW is the voluntary carbon market, where buyers increasingly favor removals that are durable, well quantified, and accompanied by development co-benefits. Two considerations bear directly on our analysis. First, integrity and measurement: because durable

removal is uncertain and partly offset by soil processes, conservative net accounting mat-
ters for what can credibly be sold (Schiedung et al., 2026; Levy et al., 2024), and registry
methodologies operationalize this through counterfactual-baseline rules – where ERW dis-
places liming, only the incremental removal is additional – and financial-additionality screens
(Suhrhoff et al., 2026). Second, additionality: a credit is meant to fund removal that would
not otherwise occur. Coupled private and public returns complicate this test in an informa-
tive way. Where yield returns alone already justify the practice, carbon credits are weakly
additional; where neither return alone suffices but their sum does, carbon finance is pivotal
and clearly additional. The spatial structure of the two returns is thus not incidental to
crediting but central to it.

What is missing. Despite rich literatures on ERW geochemistry, on the economics of
liming, and on carbon-market design, no prior work, to our knowledge, estimates the spatial
distributions of the private (yield) and public (carbon) returns to ERW side by side, char-
acterizes where each is sufficient on its own and where only their combination is, or frames
the carbon credit as the marginal financing that converts an otherwise-unprofitable yield-
improving practice into a viable one, particularly in a high-acidity, smallholder-dominated
setting like SSA. Nor has the durable-removal accounting that this coupling demands, in
which alkalinity consumed neutralizing soil acidity is not credited as exported carbon, been
applied spatially. We address these gaps.

3 Conceptual framework and modeling approach

3.1 Private and public returns

Consider a representative cropland pixel and a one-off decision to apply crushed basalt
at an agronomically calibrated rate. The per-hectare gross margin of the practice has three
economic components: an agronomic return from the yield response to acidity neutralization,

216 R_a ; a carbon return from selling the verified CO₂ removed, R_c ; and the delivered cost
 217 of the basalt, C (quarrying, grinding, transport, and spreading). All three are spatially
 218 heterogeneous. We write the combined gross margin as

$$\text{GM} = R_a + R_c - C, \quad R_c = \rho(p - m), \quad (1)$$

219 where ρ is the net CO₂ removed per hectare (t CO₂/ha, after life-cycle emissions, pedogenic-
 220 carbonate loss, and the alkalinity consumed neutralizing soil acidity; Section 3.4), p is the
 221 carbon price (\$/t CO₂), and m is the per-tonne MRV cost.

222 The novelty is not the combined margin but its *decomposition* into a private and a public
 223 test, each charged the *full* cost of the practice:

$$\text{Private-sufficient: } \text{GM}_a \equiv R_a - C > 0, \quad (2)$$

$$\text{Public-sufficient: } \text{GM}_c \equiv R_c - C > 0. \quad (3)$$

224 Equation (2) identifies land where a farmer would rationally invest on yield grounds alone,
 225 ignoring carbon; equation (3) identifies land where a carbon buyer would fund the practice
 226 on removal grounds alone, ignoring yield. Their *intersection* ($\text{GM}_a > 0 \wedge \text{GM}_c > 0$) is
 227 land where the investment is doubly justified – and, crucially, where the carbon credit is
 228 *least additional*, because the agronomic case already stands on its own. This yields a five-
 229 way typology of cropland (Figure 1): both, private-only, public-only, *combined-only* (neither
 230 stream alone covers C but together they do, $\text{GM} > 0$), and neither. The combined-only set
 231 is where carbon finance is pivotal: it is additional by construction and it is precisely the land
 232 that VCM revenue moves from unprofitable to profitable.

233 **The public envelope is a marginal-abatement-cost test.** Substituting $R_c = \rho(p - m)$
 234 into (3) and noting that the per-hectare application rate cancels between ρ and C (both scale

linearly with tonnes of rock applied), public-sufficiency reduces to

$$\underbrace{\frac{c_t}{\delta_t}}_{\text{MAC}} < p - m, \quad (4)$$

where c_t is the delivered cost per tonne of basalt and δ_t is the CO₂ removed per tonne of basalt. The public envelope is therefore exactly the land whose marginal abatement cost (MAC) lies below the net carbon price. Because δ_t – the realized removal per tonne of rock – is small and uncertain, equation (4) makes clear that the carbon case is governed first by the *CDR rate*, and only second by the credit price. This is the analytical core of the paper’s main empirical result (Section 4.3).

3.2 Data and raster modeling framework

We build on a per-pixel ($\sim 0.083^\circ$, ~ 9 km) *ex ante* model covering 44 SSA countries and 23 SPAM crops. Soil properties (pH, exchangeable acidity and bases, bulk density) come from SoilGrids (Poggio et al., 2021); harvested area, production, and yield from SPAM (Yu et al., 2020); climate from WorldClim (Fick and Hijmans, 2017); basalt source lithology from GLiM (Hartmann and Moosdorf, 2012); the 2019 motorized friction surface for haulage cost-distance from Weiss et al. (2020); and producer prices from FAOSTAT (FAO, 2023) (2016–2020 medians). Lime-equivalent requirements use the LiTAS method (Aramburu Merlos et al., 2023); the agronomic suitability response uses EcoCrop (Hijmans, 2023). Full layer provenance and constants are documented in the model repository.

3.3 Cost, carbon, and agronomic terms

Delivered cost per tonne of basalt is the sum of a quarry-gate price (\$10/t), grinding (energy use $\times$ country electricity tariff), spatial transport (travel-time $\times$ haulage rate), and spreading (\$8/t). Net CO₂ removal per tonne combines feedstock stoichiometry, a Lewis-calibrated reactive fraction (Lewis et al., 2021), a grain-size factor, a climate scalar, and a deduction for

alkalinity that re-precipitates locally as pedogenic carbonate in drier zones (Zamanian et al., 2016; Huang et al., 2024); it is then netted of grinding, transport, and spreading life-cycle emissions. The climate scalar uses an Arrhenius temperature response ($E_a = 68.8$ kJ/mol; White and Blum, 1995; Brantley et al., 2023; Deng et al., 2022), a moisture term anchored at 1,000 mm — the median precipitation across compiled ERW studies (Suhrhoﬀ et al., 2026) — and a unimodal soil-pH term peaking at pH 6, the product of dissolution kinetics that rise with acidity (Palandri and Kharaka, 2004) and an alkalinity carbon-capture eﬃciency that collapses below pH 5 (Bertagni and Porporato, 2022; Holden et al., 2024; Power et al., 2025). The grain-size lever is carried primarily as a comminution-energy trade-oﬀ (Streﬂer et al., 2018); field evidence cautions that finer grinding does not by itself guarantee proportionally faster realized weathering (Harrington et al., 2023; Suhrhoﬀ et al., 2026). The agronomic return is the modeled yield response to acidity neutralization valued at crop-speciﬁc producer prices. We adopt baseline parameters of a \$150/t CO₂ credit price — a deliberately conservative ﬂoor relative to 2024–25 market signals, where durable-CDR transactions cluster at roughly \$250–450/t CO₂ (CDR.fyi, 2025) — an MRV cost of \$20/t CO₂, at the low end of the \$15–71/t CO₂ range reported across supplier datasets (Mercer et al., 2024) and swept to \$80 in sensitivity, a 50 μ m grain size, and a 10% discount rate (a conventional project-appraisal assumption). Table 2 decomposes the delivered cost.

3.4 Net-export carbon accounting

The alkalinity released by weathering can either neutralize soil acidity or export as bicarbonate to the ocean; it cannot do both. Alkalinity that neutralizes exchangeable acidity re-releases the captured CO₂ ($\text{HCO}_3^- + \text{H}^+ \rightarrow \text{H}_2\text{O} + \text{CO}_2$, with protons supplied by Al³⁺ hydrolysis), exactly as agricultural lime does – which is why lime is not a carbon-removal technology (West and McBride, 2005; Hamilton et al., 2007). Field evidence from an acidic tropical soil conﬁrms the mechanism at ﬁrst order: over a ﬁve-year Australian basalt trial, neutralization of pre-existing acidity and non-carbonic weathering consumed nearly all of

Table 2: Delivered-cost decomposition per tonne of basalt, with the separate MRV deduction. Delivered cost = quarry-gate + grinding + transport + spreading; MRV is netted from the carbon price rather than added to delivered cost. Transport is the only spatially varying (variable) term and drives the cost heterogeneity that determines where ERW is deployable; grinding cost varies with the country electricity tariff. Life-cycle CO₂ of grinding, transport, and spreading is separately deducted from gross removal (not shown).

Component	Reference value	Type	Basis / source
Quarry-gate feed-stock	\$10/t	fixed	fines as aggregate by-product; cf. \$17.50/t crushed-stone average (U.S. Geological Survey, 2024), fines typically unpriced in production statistics (Madankan and Renforth, 2023)
Grinding (50 μ m)	19.4 kWh/t $\times$ tariff ($\approx$ \$1.9/t)	fixed	Strefler et al. (2018) energy curve
Transport (haulage)	travel-time $\times$ \$0.04/min/t	variable	30 t truck ($\approx$ \$80/h) over MAP friction surface (Weiss et al., 2020); rate within SSA corridor tariffs (Teravaninthorn and Raballand, 2009)
Spreading	\$8/t	fixed	on-farm application
Delivered cost/t	quarry + grind + transport + spread		consumed per pixel
MRV (netted separately)	\$20/t CO ₂	per t CO ₂	low end of \$15–71/t CO ₂ supplier-dataset range (Mercer et al., 2024)

the weathered alkalinity (Holden et al., 2024), and modeled carbon-capture efficiency approaches zero in strongly acid soils (Bertagni and Porporato, 2022). A recent systematic review notes that this acidity demand “is not considered in all global and regional model estimates” (Suhrhoﬀ et al., 2026). Durable removal is therefore only the alkalinity that exports *beyond* the soil’s acidity demand. We credit

$$\rho_{\text{net}} = \max(\rho_{\text{gross}} - F S, 0), \quad (5)$$

where S is the soil’s exchangeable acidity expressed as a lime requirement (t CaCO₃/ha, from the LiTAS/Kamprath calculation already in the pipeline) and $F = 0.88$ t CO₂ per t CaCO₃-eq is the CO₂ re-released per unit of acidity neutralized – fixed by the feedstock’s own

stoichiometry, not tuned. (For a silicate amendment F is twice the familiar agricultural-lime emission factor, because lime supplies half its neutralizing capacity from its own carbonate carbon whereas silicate alkalinity carries none; West and McBride, 2005; Hamilton et al., 2007.) The sink is regime-specific: the *standing* exchangeable acidity at first application, but only the annual *re-acidification* in steady state. Equation (5) is a first-order (sequential) bound: it assumes the acidity sink is filled before any alkalinity exports, whereas in reality weathering, neutralization and leaching occur together. Reactive-transport modeling moreover indicates that much of the sink acts as a *delay* rather than a permanent debit – cations retained on exchange sites can re-release and export over years to decades, depending on cation-exchange capacity, base saturation, and hydrology (Kanzaki et al., 2025) – so ρ_{net} is best read as a conservative bound on removal realized within a crediting horizon and ρ_{gross} as the long-run ceiling; the partition $\lambda \in [0, 1]$ swept in Section 4.5 spans the two, and we report both. The acidity-sink and pedogenic-carbonate deductions apply to essentially disjoint alkalinity pools – the two co-occur on only 1.3% of pixels (soils both arid and acid), where our sequential ordering over-deducts slightly and therefore conservatively. A direct corollary, central to the results, is that the agronomically optimal (lime-requirement) dose is CDR-minimal: durable removal requires applying *beyond* the acidity demand, so the private and public objectives pull the application rate in opposite directions.

3.5 Over-liming penalty

The same surplus alkalinity that exports as durable carbon also raises soil pH. Where a dose pushes pH past a crop’s optimum, yield declines through micronutrient lockout and phosphorus retrogradation. Our baseline credits the yield response to acidity relief but does not penalize this overshoot; as a refinement we evaluate each crop’s pH suitability at the *post-application* pH. We estimate that pH from the reacted alkalinity (equal to ρ_{gross} on the same F basis), which first neutralizes the soil’s exchangeable acidity and, once that demand is met, raises pH into the alkaline range buffered by the soil’s ECEC; a crop-specific EcoCrop

pH ceiling then haircuts the agronomic return above the crop’s optimum. This is a first-order refinement whose magnitude depends on the soil buffer constant and the realized reactive fraction, which we carry as a sensitivity band (Section 4.2). The penalty’s buffer and ceiling parameters are stated assumptions: the ERW literature documents the over-application risk qualitatively (Swoboda et al., 2022) but provides no quantified over-liming yield response, which is why we present a band rather than a point estimate.

3.6 Temporal regimes and allocation rules

We report three temporal accountings: *year-1* (undiscounted, agronomic return summed over the 5-year basalt residency), *NPV* at 10% (carbon revenue phased over five years and discounted; agronomic NPV over the reapplication interval), and *equilibrium* (steady-state maintenance dose). The five-year phasing tracks feedstock dissolution; realized *export* can lag dissolution where exchange sites store weathered cations – on high-CEC soils less than half of potential removal may export within the first decade (Kanzaki et al., 2025), and dissolution rates themselves decline over a decade in the field (Brantley, 2025) – a further reason the carbon case belongs on the equilibrium regime. Because durable carbon removal is intrinsically a steady-state quantity (Section 3.4), we lead both the public and the private case on the *equilibrium* regime – which places the two return streams on the same recurring, per-period footing – and report year-1 and NPV as robustness. For allocation we contrast a spatially *targeted* LiTAS dose (applied only to acid pixels above each crop’s tolerance) with a *uniform* 20 t/ha dose across all cropland, the standard convention in the ERW modeling literature.

3.7 Envelopes, aggregation, and robustness

For each pixel we compute the area-weighted (across co-located crops) agronomic return, basalt cost, and carbon revenue, and evaluate (2)–(3). We summarize each envelope by harvested area and by the aggregate private return (value of ERW-induced extra output),

public return (durable carbon revenue), and delivered cost summed over its pixels; we also report an approximate farm count (affected hectares divided by national average agricultural holding size; [Lowder et al., 2016](#)) in the country-level tables. Throughout, area shares are taken relative to the *treated* cropland—the 17.6 Mha of acid-eligible land on which the targeted dose is applied, our action set—rather than all cropland.¹ Because the per-hectare dose cancels, the public envelope, the MAC surface ((4)), and the entire parameter sweep are computed analytically from a handful of per-pixel primitives, with no model re-run. We probe robustness over delivered cost ($\times 0.5$ – 2.0), carbon price (\$50–250), yield benefit ($\times 0.5$ – 2.0), the CDR rate ($\times 0.5$ – 1.5), and the net-export partition $\lambda \in [0, 1]$ of Section 3.4, and define a pixel’s robustness score as the fraction of that grid under which it remains in the intersection. Two algebraic properties discipline how we read that grid, and are derived where they bite. Both envelope tests are homogeneous of degree zero in the delivered cost, so the five economic parameters reduce to four effective knobs for the intersection — y/c , $(p - m)/c$, r and λ (Section 4.5) — and to just two for the carbon-only envelope, which does not depend on yield (Section 4.3). Our sweep is therefore a 180-point grid over those four knobs, not a five-dimensional grid whose axes would be partly redundant. And the CDR-rate multiplier is applied to *gross* removal, before the acidity-sink deduction, because durable removal is super-linear in it. All robustness results are computed in the equilibrium steady state with net-export accounting, the same regime on which we lead the carbon case. Envelopes and the targeting geography are reproducible via the `erw-11` script; the public-envelope sensitivity via `erw-12`. Both read the decomposed cost, gross removal, and acidity-sink layers that `erw-7` writes, so no figure in this paper depends on which of the two accounting conventions is baked into the stored net-removal layer.

¹Total SSA harvested cropland is 146 Mha (SPAM). Shares of that broader base are roughly one-eighth of the treated-cropland shares reported here (17.6/146). The treated base is allocation-specific: a uniform dose is applied to all cropland the 23 SPAM crops occupy (32.6 Mha at uniform-20), so treated-cropland shares are comparable across prices within an allocation but not across allocations.

4 Results

4.1 The private, public, and intersection envelopes

Table 3 reports the three envelopes plus the combined margin in the equilibrium steady state with net-export carbon accounting. Under the targeted allocation the private (yield) envelope is the largest at 4.3 Mha, the durable public (carbon) envelope is 2.0 Mha, and the two coincide on 1.3 Mha. The combined envelope is larger still at 7.6 Mha. Decomposing targeted cropland into the typology of Section 3.1 (Figure 1): roughly 3.0 Mha is private-only, 0.6 Mha public-only, 1.3 Mha both, and **2.7 Mha is combined-only** – unprofitable on yield alone and on carbon alone, but profitable when the two revenue streams are stacked. The intersection is where a carbon purchase is *strengthened* by a co-benefit but the credit is least additional; the combined-only band is where carbon finance is *pivotal* – additional by construction, and precisely the land that VCM revenue moves from unprofitable to profitable. This is the paper’s central economic result: across most ERW-viable land in SSA, voluntary carbon finance is not a windfall on an already-profitable practice but the marginal revenue that makes an otherwise unprofitable agronomic investment pencil out.

The allocation comparison makes the agronomy–carbon dose trade-off concrete (Table 3). Moving from the targeted (lime-requirement) dose up the uniform ladder, the private envelope shrinks monotonically (4.3, 3.5, 1.3 and 0.5 Mha at targeted, 10, 20 and 50 t/ha), because a fixed dose increasingly wastes rock on low-response soils, while the durable public envelope and especially its carbon yield grow (public area 2.0, 3.6, 4.0 and 4.2 Mha; durable CDR 5.6, 8.8, 20 and 56 Mt), because applying beyond the acidity demand leaves more alkalinity free to export. The combined footprint peaks near the agronomic dose (7.6–8.3 Mha at targeted and uniform-10) and the carbon yield at uniform-50, the latter at far higher cost. This is the dose-design corollary of Section 3.4: maximizing private return points to the lime requirement, maximizing durable removal points to over-application, and the two coincide only where the surplus pays for itself. We therefore report the full uniform ladder alongside

Table 3: Private, public, and combined return envelopes in the equilibrium steady state with net-export carbon accounting, by allocation rule (targeted; uniform 10/20/50 t/ha). *Area* is harvested hectares (Mha). *Private* is the value of additional crop output from ERW; *Public* is durable net-export CDR valued at \$150/t CO₂ net of the \$20/t CO₂ MRV cost; *Cost* is delivered basalt (quarry, grinding, transport, spreading). Monetary totals are \$M per maintenance cycle, summed over the pixels in each envelope; combined gross margin = Private + Public – Cost. *CDR* is durable net-export removal (Mt).

Allocation	Envelope	Area	Private (\$M)	Public (\$M)	Cost (\$M)	CDR (Mt)
Targeted	Private	4.31	3,496	1,092	1,743	8.4
Targeted	Public	1.95	1,400	730	489	5.6
Targeted	Intersection	1.32	1,317	556	362	4.3
Targeted	Combined > 0	7.61	4,553	1,810	3,145	13.9
Uniform 10	Private	3.52	3,277	846	1,432	6.5
Uniform 10	Public	3.61	1,431	1,138	808	8.8
Uniform 10	Intersection	1.21	1,280	406	275	3.1
Uniform 10	Combined > 0	8.27	4,214	2,139	3,096	16.5
Uniform 20	Private	1.29	1,876	766	849	5.9
Uniform 20	Public	3.95	1,477	2,649	1,806	20.4
Uniform 20	Intersection	0.75	1,140	547	340	4.2
Uniform 20	Combined > 0	6.38	3,301	3,800	4,045	29.2
Uniform 50	Private	0.45	997	822	519	6.3
Uniform 50	Public	4.20	1,504	7,267	4,864	55.9
Uniform 50	Intersection	0.41	893	778	442	6.0
Uniform 50	Combined > 0	4.70	1,936	7,927	5,766	61.0

the targeted dose, which we retain as the agronomically motivated default.

4.2 Durable carbon and the agronomy–carbon trade-off

The carbon figures above already net out the alkalinity consumed neutralizing soil acidity (Section 3.4), and that deduction is first-order, not cosmetic. At first application the standing acidity absorbs most of the weathered alkalinity, so under a third of gross removal is durably exported (targeted: 99 Mt gross → 29 Mt net; uniform-20: 135 → 38 Mt). In the equilibrium steady state, where the maintenance dose offsets only annual re-acidification, most removal survives (targeted: 32 → 25 Mt, 77%; uniform-20: 135 → 121 Mt, 90%). Durable removal from ERW on acid soils is thus largely a *steady-state* phenomenon; the first application is, to first order, liming with a small carbon tail – which is why we evaluate the public case at equilibrium.

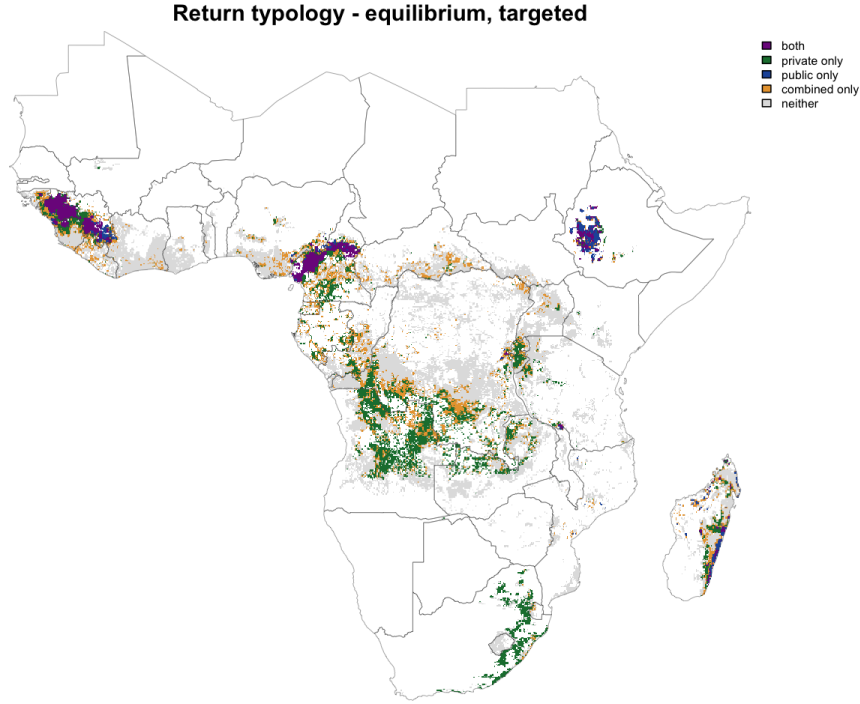


Figure 1: Return typology of SSA cropland in the equilibrium steady state with net-export carbon accounting, targeted allocation. Purple = both private and public returns independently sufficient (intersection); green = private-only; blue = public-only; orange = combined-only (viable only when both streams are stacked); grey = neither.

The same accounting resolves how returns vary with soil pH. Both the yield benefit and the gross weathering rate are largest on the most acidic soils, so a naive model would place *both* returns on the same low-pH land. Once the acidity sink is deducted, the durable-carbon return shifts toward *less* acidic soils – where less alkalinity is spent on neutralization and more exports – so the private and public returns become spatially complementary (Figure 3). The corollary for deployment design: because the lime-requirement dose is sized to the acidity demand, it is carbon-minimal; maximizing durable removal requires applying *beyond* agronomic need, so the two objectives imply different optimal rates and the carbon payment must cover the surplus rock that yields no extra harvest.

Over-application also carries an agronomic cost that the base model omits: pushing soil pH past the crop optimum reduces yield (Section 3.5). Charging this over-liming penalty

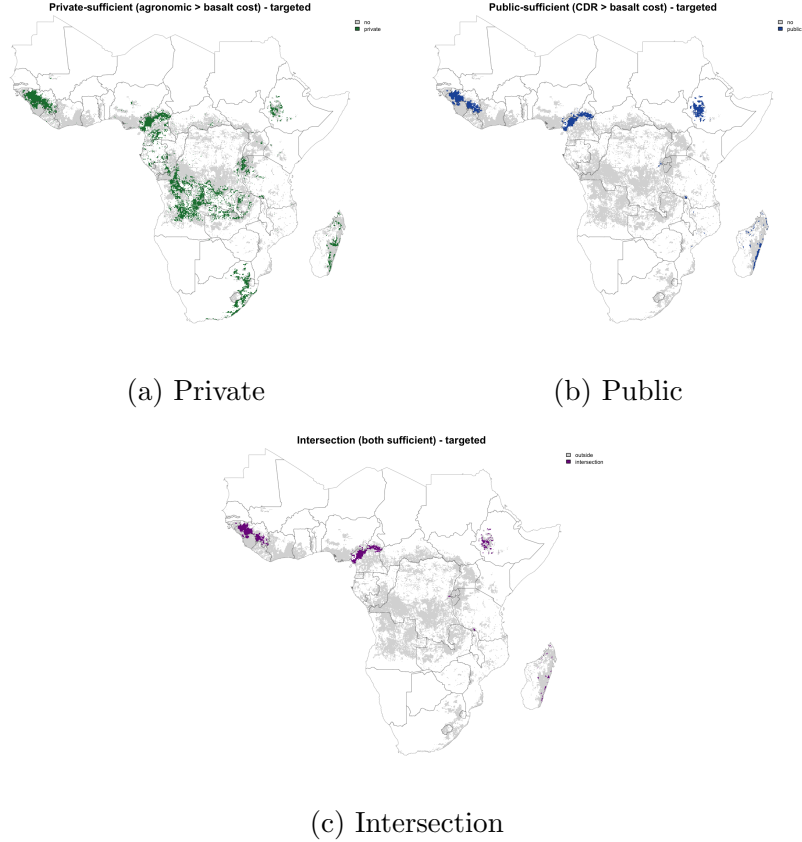
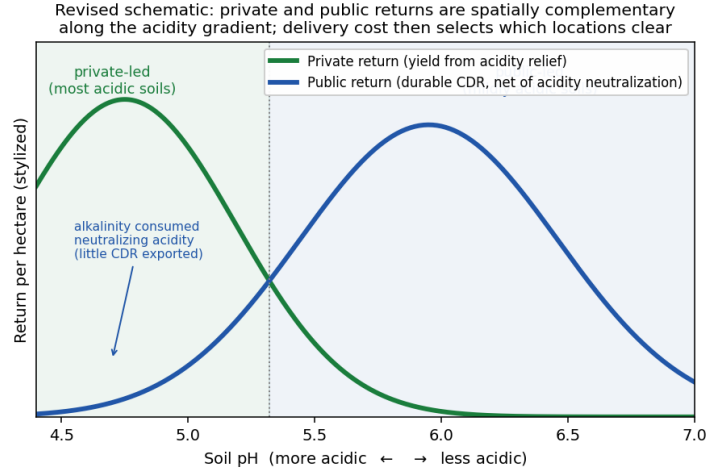


Figure 2: The three envelopes mapped separately (equilibrium steady state, net-export carbon accounting, targeted). Private: agronomic return exceeds delivered cost. Public: durable net-export carbon revenue at \$150/t CO₂ exceeds delivered cost. Intersection: both hold.

leaves the private return essentially unchanged at the targeted dose but erodes it at high uniform rates, and it does so with sharply increasing uncertainty (Figure 4). Centrally, the penalty reduces the aggregate private return by about 7% at the targeted dose, 6% at uniform-10, 6% at uniform-20, and 20% at uniform-50; but the plausible range widens from a tight 6–10% at the agronomic dose to fully 3–99% at uniform-50, depending on soil buffering and realized reactivity. The direction is robust while the high-rate magnitude is not: blanket high rates trade a bounded carbon gain for a large and poorly-bounded yield risk, whereas the targeted dose avoids the exposure. This is the third margin – alongside rising cost and the acidity-sink deduction on CDR – on which over-application is disfavored, and it completes the case for targeting plus modest over-application rather than uniformly



Realized returns are additionally gated by the delivery-cost gradient (distance to basalt): within each band, only low-cost, basalt-proximate pixels clear.

Figure 3: Private and public returns are spatially complementary along the soil-pH gradient once durable (net-export) carbon is credited: private (yield) returns peak on the most acidic soils; public returns peak on mildly acidic soils where weathered alkalinity exports rather than neutralizing acidity. Realized returns are then further selected by the delivery-cost gradient.

heavy rates.

4.3 What binds the public (carbon) envelope

Because ERW investment interest concentrates on the carbon case, we diagnose it directly through the MAC test of equation (4), on durable net-export removal. We lead this subsection on the *uniform-20* allocation rather than the targeted dose. The reason is structural, and follows from Section 3.4: the acidity sink is a soil property and is nearly invariant to the dose (0.42 t CO₂/ha of alkalinity consumed at both uniform-20 and targeted), while gross removal scales with the rock applied. The lime-requirement dose is therefore *CDR-minimal by construction* – almost all of its alkalinity is spent neutralizing the acidity it was sized to neutralize, and little exports. Diagnosing the carbon case on the targeted dose understates the durable public envelope by a factor of 2.0 in area and 3.6 in removal (2.0 versus 4.0 Mha; 5.6 versus 20.4 Mt). The targeted dose remains our lead for the private and intersection geography (Sections 4.1 and 4.5), where it is the agronomically motivated default.



Figure 4: Over-liming penalty on the aggregate private return across the application-rate ladder: no-penalty base, central estimate, and the sensitivity band spanning soil-buffer and realized-reactivity assumptions. The band is tight at the agronomic (targeted) dose and fans out at high uniform rates.

In the equilibrium steady state the area-weighted MAC under uniform-20 has a median of **\$276/ t CO₂** (Figure 5a; \$330 under the targeted dose), against a net credit price of only \$130 (\$150 less \$20 MRV); only the cheapest fraction of land clears, which is why the durable public envelope is 4.0 Mha. The contrast with first application is stark: under net-export accounting the NPV median MAC rises to roughly \$840/ t CO₂ and the carbon case all but disappears, because the standing acidity must be paid down before any carbon exports – again locating durable removal in the steady state.

Our MAC levels sit above published cost syntheses, and the gap is informative rather than anomalous. A harmonized compilation across 25 ERW costing studies reports a scenario-level median near \$195/ t CO₂ (interquartile range roughly \$137–276), with Global South locations toward the lower end (Suhrhoff et al., 2026). Our uniform-20 median (\$276) lands at the upper edge of that interquartile range and the targeted median (\$330) just above it. Decomposing $MAC = c_t / \delta_t$ locates the residual difference: our delivered cost per tonne of *rock* is in line with that literature, but our denominator credits only durable net-export removal – after the acidity-sink, pedogenic-carbonate, and life-cycle deductions – whereas

most published estimates credit gross or potential removal, which the same review identifies as a pervasive comparability problem. On a gross basis our MAC distribution falls well inside the published range. The comparison is a statement about accounting stringency, not about SSA logistics.

The durable public envelope of 4.0 Mha is 12.1% of the cropland treated under a uniform-20 dose (2.7% of all SSA cropland), and its size is governed by the carbon price rather than by any kinetic assumption: carbon-alone deployable area rises steeply as the price climbs, from 4.3% of treated cropland at \$100/t CO₂ to 12.1% at today’s \$150, 71% at \$350, and 86% at \$500 (Table 4). Notably the share of treated land that clears is almost identical to the targeted case (11.1% at \$150): uniform-20 widens the carbon envelope by treating more land and by exporting more of each tonne’s alkalinity, not by clearing a larger fraction of what it treats. Two implications follow. First, the physical removal potential is large but its *economic* viability under carbon revenue alone is price-limited: at present prices durable CDR pays for its delivered cost on under a tenth of the ERW-relevant cropland. Second—and this is the crux of the argument—the fact that carbon revenue rarely clears on its own is precisely why the private co-benefit matters. Land that carbon alone cannot reach at \$150 becomes viable once the yield return is stacked on top (the combined-only band, Section 4.1); the co-benefit is not a nicety but the mechanism that mobilizes voluntary carbon finance across the region. And because the marginal abatement cost is the ratio of delivered cost to CO₂ removed per tonne, the two routes to broadening carbon-alone viability are a higher realized CDR rate and cheaper delivery, both of which lower it.

One caveat on the arithmetic of that last sentence. The public test $\delta_t(p - m) > c_t$ is homogeneous of degree zero in the pair $((p - m), c_t)$: the carbon price, the MRV cost, and the delivered cost enter the public envelope *only* through the ratio $(p - m)/c_t$. They are one lever, not three, and a sensitivity analysis that perturbs them separately reports the widths of its own bands rather than three distinct uncertainties. The genuinely distinct levers on the carbon case are that price–cost ratio and the realized removal rate δ_t .

Table 4: Carbon-alone deployability is price-conditional. Public-sufficient area (durable net-export CDR revenue exceeds delivered cost) in the equilibrium steady state, *uniform-20* allocation, across VCM carbon prices. Shares are of the 32.6 Mha of cropland treated under a uniform-20 dose; the all-cropland share (of 146 Mha) is shown for reference. The treated base is allocation-specific: the targeted dose is applied only to the 17.6 Mha of acid-eligible cropland, so its percentage shares are not comparable in levels to the uniform ladder. Targeted values are given in brackets for reference.

VCM price (\$/t CO ₂)	Public area (Mha)	% treated	% all cropland	Durable CDR (Mt)
100	1.40 [0.49]	4.3 [2.8]	1.0	7.8 [2.2]
150	3.95 [1.95]	12.1 [11.1]	2.7	20.4 [5.6]
250	7.00 [2.88]	21.5 [16.4]	4.8	33.3 [7.0]
350	23.06 [8.35]	70.7 [47.5]	15.8	100.6 [17.5]
500	27.98 [13.42]	85.8 [76.4]	19.2	115.6 [23.3]

The binding factor is the realized CO₂ removed per tonne of rock, δ_t (here on the order of 0.1 to 0.25 tonnes CO₂ per tonne of rock), not the credit price or logistics. Figure 6 makes this quantitative, now on the headline equilibrium net-export basis: doubling δ_t increases the public envelope nearly six-fold (4.0 to 22.5 Mha), far outstripping a 33% higher carbon price (+34%) or an equivalent delivered-cost reduction. Two qualifications keep this honest. First, δ_t must be scaled where it physically acts – on *gross* removal, before the acidity sink is deducted. Because the sink and the lifecycle term are fixed subtrahends inside a $\max(\cdot, 0)$, durable removal is super-linear in δ_t ; scaling the already-deducted quantity instead would report only 8.4 Mha at $2\delta_t$ rather than 22.5 Mha, understating the very lever the argument rests on. Second, the dominance is a property of *large* moves in δ_t , not of local slopes: at small steps the two levers are comparable (a 25% higher δ_t adds 1.1 Mha; a price move to \$200 adds 1.4 Mha), though across the full one-at-a-time sweep the δ_t swing (6.4 Mha) now exceeds that of the price–cost ratio (5.6 Mha). The super-linearity bites harder on the targeted dose, where the acidity sink absorbs 22% of gross removal rather than uniform-20’s 8%. δ_t is set by the reactive fraction, grain size, the climate factor, and pedogenic-carbonate loss – the parameters that current process knowledge and MRV resolve least well. The economic and the scientific frontier therefore coincide: the largest lever on the carbon case is also the least-resolved quantity.

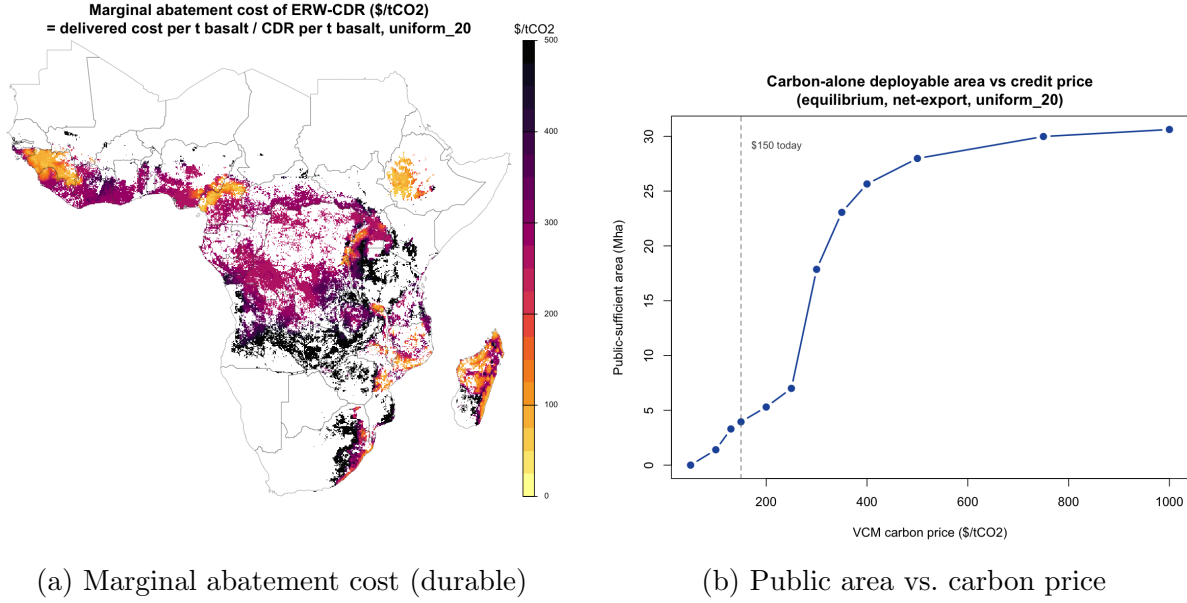


Figure 5: (a) Marginal abatement cost of durable net-export ERW-CDR (\$/ t CO₂, capped at 500; equilibrium steady state, uniform-20 allocation). (b) Public-sufficient area as a function of the VCM carbon price, on the same equilibrium net-export basis; the schedule is steeply convex around current prices. Both panels are led on uniform-20 because the targeted lime-requirement dose is CDR-minimal by construction (Section 4.3).

4.4 Cost-side levers: at-scale solar haulage

Transport is the largest single component of delivered cost ($\sim 55\%$ on average, median $\sim 66\%$) and is wholly variable in distance – consistent with life-cycle and techno-economic assessments in which transport dominates both cost and embodied emissions under long hauls (Zhang et al., 2023; Kroeger et al., 2026). We evaluate an at-scale solar-electric haulage scenario that displaces the fuel/energy portion of trucking cost. Grounding the energy share in trucking-cost data and the diesel-to-solar-electric energy gap implies a feasible $\sim 25\%$ reduction in the transport term (range 15–40%), i.e. $\sim 14\%$ off average delivered cost; the reduction is a scenario assumption about a technology transition, not an observed deployment. Applied to the headline grid (equilibrium steady state, net-export accounting, targeted allocation, so the comparison is against the headline envelopes), this widens the *private* envelope materially — from 4.31 to 5.12 Mha at -25% and to 5.80 Mha at -40% — but the durable public envelope only slightly, from 1.95 to 2.07 and 2.16 Mha. The intersection tracks the private

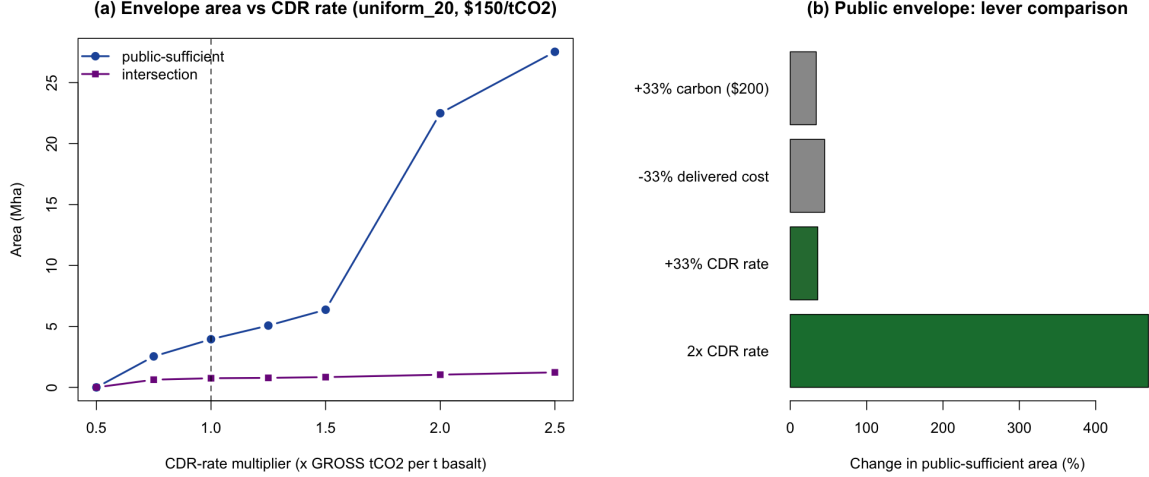


Figure 6: The CDR rate is the binding lever for the public envelope (equilibrium, net-export, uniform-20). (a) Public and intersection area versus the realized CDR-rate multiplier, applied to *gross* removal before the acidity-sink deduction. (b) Change in public-sufficient area under alternative levers: doubling the CDR rate dominates plausible carbon-price or delivered-cost moves, though at the $\pm 33\%$ step the cost lever is marginally the stronger. Note that the carbon-price and delivered-cost bars are the same lever at different step sizes, since the public envelope depends on them only through $(p - m)/c_t$.

side (1.32 to 1.41 and 1.49 Mha). This confirms the asymmetry implied by equation (4): a cost reduction relaxes the private test dollar-for-dollar, but relaxes the public test only where it drags a pixel's marginal abatement cost below the net credit price, which happens on comparatively little land. Carbon price and CDR rate move the public envelope; logistics cost moves the private one. The two policy levers are thus complementary and target different margins. (This scenario holds transport LCA emissions fixed; because solar haulage would also cut them, the public-side gain reported here is a conservative lower bound.)

4.5 Targeting and robustness

We compute the robustness of the doubly-justified geography on the same equilibrium steady state with net-export accounting on which the carbon case is led. The intersection test is homogeneous of degree zero in the delivered cost: dividing both envelope conditions by the cost multiplier c leaves private : $a_i(y/c) > c_i$ and public : $\delta_i(r, \lambda) ((p - m)/c) > c_i$, so the intersection depends on the five economic parameters only through *four* effective knobs: the

yield/cost ratio y/c , the net-price/cost ratio $(p - m)/c$, the CDR-rate multiplier r applied to gross removal, and the net-export partition λ . We sweep a $5 \times 4 \times 3 \times 3 = 180$ -point grid over these and define a pixel’s robustness score as the fraction of the grid under which it remains in the intersection (Figure 7a).

No pixel survives more than 61% of this deliberately wide grid: the doubly-justified case is inherently parameter-sensitive. A core-targeting tier (in the intersection under at least half the grid) covers 0.65 Mha, ~ 0.38 M farms and $\sim \$0.60$ B of crop value, and a broader candidate tier (at least one-third) covers 1.22 Mha, ~ 0.76 M farms and $\sim \$1.06$ B (Figure 7b). Both concentrate sharply in the Cameroon highlands (0.46 Mha core, 0.61 Mha candidate) and the Guinean highlands (0.16 and 0.38 Mha), extending thinly into the Ethiopian highlands, Tanzania and Sierra Leone – areas where acidic soils, a favorable weathering climate, basalt proximity, and adequate crop prices coincide. Four-fifths of the candidate tier lies in two countries.

A one-at-a-time tornado around the central case (baseline intersection 1.32 Mha) ranks the four knobs: the CDR rate swings the intersection by 1.72 Mha, the net-price/cost ratio by 1.73 Mha, the yield/cost ratio by 0.96 Mha, and λ by only 0.22 Mha. Two readings matter. First, on the intersection the CDR rate and the price–cost ratio are effectively *tied*; the CDR rate’s clear lead over price appears in the carbon-only envelope (Section 4.3), not here, because the intersection is jointly constrained by the agronomic test that the CDR rate does not enter. Second, earlier drafts reported delivered cost as the dominant lever by a wide margin. On this basis its raw swing is indeed the largest (2.78 Mha), but that is an artifact of aggregation rather than a separate mechanism: cost is the only parameter that moves *both* ratios. Holding y/c fixed, its public channel swings the intersection by 1.94 Mha; holding $(p - m)/c$ fixed, its private channel swings it by 0.96 Mha – exactly the yield swing, as it must be, since yield moves y/c and nothing else. Delivered cost matters because it is levered twice, not because logistics dominates weathering. The reassuring corollary stands: the yield benefit, the most uncertain input given the thin field-trial base, is among the least

Table 5: The four return envelopes under three coherent parameter bundles (targeted allocation, equilibrium steady state, net-export accounting; area in Mha). Conservative sets every lever to its ERW-unfavorable end ($p = \$100$, MRV \$40, delivered cost $\times 1.5$, CDR rate $\times 0.5$, yield $\times 0.75$); Optimistic sets every lever favorable (\$250, \$10, $\times 0.75$, $\times 1.5$, $\times 1.5$); Central is the baseline of Table 3.

Envelope	Conservative	Central	Optimistic
Private	1.36	4.31	8.73
Public	0.00	1.95	15.64
Intersection	0.00	1.32	8.27
Combined > 0	1.56	7.61	16.81

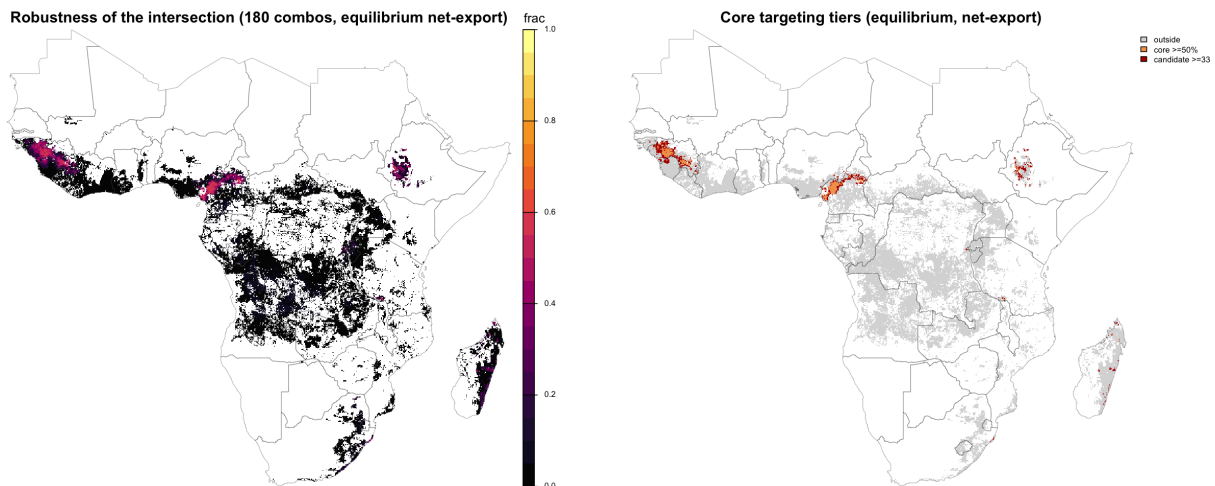
influential, and λ – the most-scrutinized modeling choice – is the least influential of all.

To make the range concrete for a non-technical reader, we also stack the five levers into three coherent bundles (Table 5): a Conservative case with every lever at its ERW-unfavorable end, the Central case of Table 3, and an Optimistic case with every lever favorable. The qualitative story is robust while the magnitudes are not. The private envelope never collapses — it holds at 1.4 Mha even under the Conservative bundle — and the combined footprint is always the largest of the four. The carbon-only case is far more fragile: the public envelope vanishes entirely under the Conservative bundle (a \$100 price, \$40 MRV, a halved CDR rate, and $1.5\times$ delivered cost together clear no land against the resulting net price) and swells past 15 Mha under the Optimistic one — an asymmetry that is itself the point, since the upside is driven by the CDR rate acting on gross removal, where it is super-linear. This is the honest headline: the *ordering* of the envelopes and the pivotal role of the combined-only band survive the whole grid, but the absolute carbon geography is contingent, and most contingent on the CDR rate.

The carbon case behaves very differently across regimes once durable removal is credited (Table 6). Under gross accounting the public envelope looks broadly similar at first application and in steady state; under net-export accounting it nearly vanishes at first application (NPV public $2.46 \rightarrow 0.45$ Mha), because the standing acidity consumes the alkalinity, and is restored only in the equilibrium steady state ($2.46 \rightarrow 1.95$ Mha), where the maintenance

Table 6: Public envelope and total credited CDR (targeted), under gross versus net-export accounting, at first application (NPV) and in steady state (equilibrium). Net-export accounting collapses the first-application carbon case and locates durable removal in the steady state.

Accounting	Public, NPV (Mha)	Public, equil. (Mha)	CDR, NPV (Mt)	CDR, equil. (Mt)
Gross	2.46	2.46	99	32
Net-export	0.45	1.95	29	25



(a) Per-pixel robustness score

(b) Core-targeting tiers

Figure 7: Robustness of the doubly-justified geography, computed on the headline equilibrium steady state with net-export accounting. (a) Fraction of the 180-point grid over the four effective knobs (y/c , $(p-m)/c$, r , λ) under which a pixel remains in the intersection; no pixel exceeds 0.61. (b) Red = robust core (intersection under $\geq 50\%$ of the grid); orange = candidate tier ($\geq 33\%$). The core concentrates where soil acidity, weathering climate, basalt access, and crop value coincide.

dose offsets only annual re-acidification. The private envelope, which does not depend on the carbon accounting, is stable across regimes. This regime-dependence is precisely why we lead the carbon case on equilibrium.

5 Discussion

Three implications follow for the design of ERW programs and for the agricultural economics of carbon-coupled practices more generally.

Carbon finance is the marginal, not the windfall, revenue. The dominance of the combined-only band means that on most ERW-viable SSA cropland the agronomic return and the carbon return are each individually insufficient to cover the upfront cost of crushed basalt, but their sum is not. Voluntary carbon revenue is therefore best understood as the financing that crosses the adoption threshold for a yield-enhancing practice that farmers would otherwise forgo – a framing closer to a cost-share or input subsidy than to a pure carbon transaction. This both strengthens the additionality claim on that land (the practice does not happen without the payment) and reframes the distributional question: the credit buyer effectively subsidizes a private yield gain, and program design must decide how that joint surplus is split between buyer and farmer.

The carbon case is CDR-quantification risk, not price risk. Equation (4) and Figure 6 show that the public envelope hinges on the realized removal per tonne of rock, the largest lever on the carbon-only case and the least-resolved scientific quantity. The claim should be stated with its margin. On the carbon-only envelope the CDR rate leads the price–cost ratio by 6.4 to 5.6 Mha of swing, and on the doubly-justified intersection the two are effectively tied (1.72 versus 1.73 Mha); at small perturbations the two levers are comparable. What distinguishes δ_t is the *shape* of its response, not its local slope: because the acidity sink and lifecycle emissions are fixed subtrahends, durable removal is super-linear in δ_t , so large improvements in the realized rate compound in a way that price and cost improvements do not. Doubling δ_t increases the public envelope nearly six-fold; no plausible price or cost move comes close. Together with the observation that price, MRV, and delivered cost are one lever rather than three, this relocates the central risk for investors from the carbon market (price) to measurement, reporting, and verification (quantity and durability). It also reframes MRV: the \$20/t CO₂ MRV cost is not merely a deduction but the mechanism that converts an uncertain δ_t into a sellable, high-integrity tonne. Investment in cheaper, tighter measurement raises the *effective* CDR rate that can be monetized – exactly where

the leverage lies.

Targeting and sequencing. The robustness analysis points to a small core geography that is attractive across a wide range of assumptions, and a broader candidate tier for early deployment. Because cost-side and revenue-side levers move different envelopes, an efficient program pairs logistics investment (basalt siting, low-carbon haulage) where the private case is close, with carbon-price or CDR-rate improvements where the public case is close.

Where field data would most improve the model. The robustness analysis doubles as a data-collection agenda, and answers the sixth question the introduction poses. The single input that most moves both the carbon-only and the doubly-justified geography is the realized CO₂ removed per tonne of rock, δ_t , which the model carries as a climate- and pH-scaled reactive fraction calibrated largely on temperate soils. The priority measurements, in order of leverage, are therefore: (i) tropical, smallholder-system CDR rates that pin down δ_t and its climate scaling — the top lever on the public envelope and, jointly with the price–cost ratio, on the doubly-justified area; (ii) the pedogenic-carbonate fraction in semi-arid zones, which the aridity-band deduction only proxies and which enters δ_t directly; and (iii) delivered-cost calibration near candidate-tier basalt sources — the levered price–cost ratio — which moves the private envelope directly and the intersection through both of its channels. Notably the yield response, the input with the thinnest evidence base (a prior-dominated hierarchical fit, via [Bürkner, 2017](#), on roughly a dozen trials), is where additional data would change the targeting map *least*, because the private envelope is robust to it. Field effort is best spent on the carbon quantity rather than the agronomic one, and concentrated in the candidate-tier geography, where a measurement is most likely to change a deployment decision rather than confirm one. The economic frontier and the scientific frontier are, once again, the same quantity.

Limitations. The year-1 accounting is the most optimistic and the NPV the most conservative; we report both. Farm counts use national average holding sizes and are order-of-magnitude. The agronomic response and the climate and pedogenic-carbonate terms remain empirical proxies; the CDR-rate sweep is our stand-in for that structural uncertainty, and tighter field calibration is the clear priority. We also omit downstream transport losses – river degassing and in-stream carbonate precipitation, and air–sea equilibration losses – which published estimates place at roughly 15–20% of exported removal combined (Renforth and Henderson, 2017; Zhang et al., 2025), with the low-latitude mixing term at the higher end of that range (Bertagni and Porporato, 2022); our reported removals are gross of these, so the omission biases the carbon case modestly upward. Conversely, cation-exchange dynamics, which delay rather than destroy export on high-CEC soils (Kanzaki et al., 2025), are absorbed only informally by the λ partition, and omitted N₂O effects likely run in ERW’s favor, as most studies report emission decreases (Suhrhoﬀ et al., 2026; Kroeger et al., 2026). We do not model tenure, who finances the upfront cost, soil compaction, or country-level credit eligibility (Article 6), each of which would refine the deployable geography.

6 Conclusion

Enhanced Rock Weathering is unusual among carbon-removal options in paying a farmer twice: once in yield and once in carbon. Mapping those two returns separately across Sub-Saharan Africa shows that they rarely each suffice on their own – but that together they make a meaningful area of cropland profitable that neither would alone. The policy object of interest is therefore the *combined-only* land, where voluntary carbon finance is the marginal revenue that unlocks a yield-improving investment, and the binding technical constraint is how much CO₂ a tonne of rock verifiably removes. Treating ERW as a coupled agronomic-and-carbon proposition, rather than as a pure CDR supply curve, changes both where one would deploy it first and what one would invest in to expand it.

Data and code availability

All input layers are public. The analysis pipeline, the envelope and robustness scripts (`erw-1-erw-11`), and the code that reproduces the figures and tables reported here are available in the project repository `erw-exante-v2` (<https://github.com/chamb244/erw-exante-v2>). Large input rasters are distributed separately on request; every raster’s public source is documented in the repository.

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